# ON AN AFFINE SPACE OVER A NOETHERIAN BASE EVERY QUASI-COHERENT MODULE IS A FILTERED COLIMIT OF FINITELY PRESENTED MODULES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

*Editorial note.* Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

*Allowed standard prerequisites (editorial).*

- Every quasi-coherent module on a Noetherian space is a filtered colimit of coherent modules. Same chapter spaces-cohomology.tex, label lemma-directed-colimit-coherent (invoked at proof line 2909). It writes $f_*\mathcal{F} = \colim \mathcal{H}_i$ with $\mathcal{H}_i$ coherent, which is the system of finitely presented modules that is pulled back.
- Coherence on a Noetherian affine base is finite presentation. Same chapter spaces-cohomology.tex, label lemma-coherent-Noetherian (invoked at proof line 2910). It upgrades the coherent modules $\mathcal{H}_i$ to modules of finite presentation, so that their pullbacks are finitely presented as well.
- Base change of pushforwards and pullbacks along an etale cover, which reduces the surjectivity claim to schemes. Properties of Spaces, label spaces-properties-lemma-pushforward-etale-base-change-modules (invoked at proof line 2924). With $U = X \times_Y V$ it identifies the restriction of $f^*f_*\mathcal{F} \to \mathcal{F}$ with the corresponding map for the affine morphism of schemes $f' : U \to V$, where surjectivity is known.
- A morphism between filtered colimits of finitely presented modules factors through a stage. Same chapter spaces-cohomology.tex, label lemma-finite-presentation-quasi-compact-colimit (invoked at proof line 2953). For every $j$ it produces a stage $i$ and a morphism $\alpha : \mathcal{G}_j \to \mathcal{H}_i$ making the obvious square commute, which is what builds the directed set of triples.

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*Date*: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 07UB: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2094–2094. Navigation pointer for \ref{lemma-coherent-Noetherian} used inside the counted span (line 2910); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 07UV: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2866–2866. Navigation pointer for \ref{lemma-directed-colimit-coherent} used inside the counted span (line 2909); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 07U7: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 735–735. Navigation pointer for \ref{lemma-finite-presentation-quasi-compact-colimit} used inside the counted span (line 2953); the argument that consumes it is retained in the span.

*Source reference.* Original lemma 03LX: sources/stacks-chow/tags.txt, source lines 4729–4729. Navigation pointer for \ref{spaces-properties-lemma-pushforward-etale-base-change-modules} used inside the counted span (line 2924); the argument that consumes it is retained in the span.

*Source reference.* Original section 05VR: sources/stacks-chow/tags.txt, source lines 7560–7560. Navigation pointer for \ref{spaces-properties-section-properties-modules} used inside the counted span (line 2914); the argument that consumes it is retained in the span.

**Lemma 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be an affine morphism of algebraic spaces over $S$ with $Y$ Noetherian. Then every quasi-coherent $\mathcal{O}_X$-module is a filtered colimit of finitely presented $\mathcal{O}_X$-modules.*

*Proof.* Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Write $f_*\mathcal{F} = \operatorname{colim} \mathcal{H}_i$ with $\mathcal{H}_i$ a coherent $\mathcal{O}_Y$-module, see Lemma 07UV. By Lemma 07UB the modules $\mathcal{H}_i$ are $\mathcal{O}_Y$-modules of finite presentation. Hence $f^*\mathcal{H}_i$ is an $\mathcal{O}_X$-module of finite presentation, see Properties of Spaces, Section 05VR. We claim the map

$$\operatorname{colim} f^*\mathcal{H}_i = f^*f_*\mathcal{F} \to \mathcal{F}$$

is surjective as $f$ is assumed affine, Namely, choose a scheme $V$ and a surjective étale morphism $V \to Y$. Set $U = X \times_Y V$. Then $U$ is a scheme, $f' : U \to V$ is affine, and $U \to X$ is surjective étale. By Properties of Spaces, Lemma 03LX we see that $f'_*(\mathcal{F}|_U) = f_*\mathcal{F}|_V$ and similarly for pullbacks. Thus the restriction of $f^*f_*\mathcal{F} \to \mathcal{F}$ to $U$ is the map

$$f^*f_*\mathcal{F}|_U = (f')^*(f_*\mathcal{F})|_V) = (f')^*f'_*(\mathcal{F}|_U) \to \mathcal{F}|_U$$

which is surjective as $f'$ is an affine morphism of schemes. Hence the claim holds.

We conclude that every quasi-coherent module on $X$ is a quotient of a filtered colimit of finitely presented modules. In particular, we see that $\mathcal{F}$ is a cokernel of a map

$$\operatorname{colim}_{j\in J} \mathcal{G}_j \longrightarrow \operatorname{colim}_{i\in I} \mathcal{H}_i$$

with $\mathcal{G}_j$ and $\mathcal{H}_i$ finitely presented. Note that for every $j \in I$ there exist $i \in I$ and a morphism $\alpha : \mathcal{G}_j \to \mathcal{H}_i$ such that

$$\begin{array}{ccc} \mathcal{G}_j & \xrightarrow{\alpha} & \mathcal{H}_i \\ \downarrow & & \downarrow \\ \operatorname{colim}_{j\in J} \mathcal{G}_j & \longrightarrow & \operatorname{colim}_{i\in I} \mathcal{H}_i \end{array}$$

commutes, see Lemma 07U7. In this situation $\mathrm{Coker}(\alpha)$ is a finitely presented $\mathcal{O}_X$-module which comes endowed with a map $\mathrm{Coker}(\alpha) \to \mathcal{F}$. Consider the set $K$ of triples $(i, j, \alpha)$ as above. We say that $(i, j, \alpha) \leq (i', j', \alpha')$ if and only if $i \leq i'$, $j \leq j'$, and the diagram

$$\begin{array}{ccc} \mathcal{G}_j & \xrightarrow{\alpha} & \mathcal{H}_i \\ \downarrow & & \downarrow \\ \mathcal{G}_{j'} & \xrightarrow{\alpha'} & \mathcal{H}_{i'} \end{array}$$

commutes. It follows from the above that $K$ is a directed partially ordered set,

$$\mathcal{F} = \mathrm{colim}_{(i,j,\alpha) \in K} \mathrm{Coker}(\alpha),$$

and we win. □